

Lambda: A 4 mm² Open-Source Transformer Inference ASIC

with Compressed-Domain Attention and Adaptive-Precision KV

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Abstract

We present Lambda, a 4 mm² standalone transformer-decode ASIC at TSMC N16FFC, designed for tape-out via the IMEC / Europractice mini@sic 2.0 academic shuttle program. Lambda runs 3–5 B-parameter W4A8 transformer models—Llama-3.2-3B, Qwen2.5-3B, Phi-3.5-mini—at 6.3–7.7 tokens/s decode in a 2.6 W typical envelope, on a single LPDDR5X-8533 x16 channel sustaining 12 GB/s. Three architectural contributions distinguish the chip. (1) We implement a TurboQuant-style rotation-codebook KV compressor in silicon at 4.0 effective bits per element (4.0× vs. FP16), via a 16-point Walsh–Hadamard butterfly and 8-centroid Lloyd–Max codebook—a zero-multiplier compression engine occupying 0.08 mm²; quality parity at 4.0 bpe is a design target pending a pre-HLS golden-model evaluation, since the published TurboQuant results use full-head-dimension rotation at 3.5 bpe. (2) We perform compressed-domain attention scoring directly on 3-bit codebook indices (INT8 × INT3 with an INT24 reduction along the head dimension), eliminating the need for an FP16 multiplier fabric. (3) We introduce a Token Importance Unit (TIU): a 0.03 mm² block that accumulates per-block attention entropy and drives both heavy-hitter eviction and adaptive per-block precision in the KV cache. The complete chip integrates seven functional units under an ACU/MSC/LSU/TIU/HIF top-level hierarchy and presents a PCIe Gen3 x1 host interface on an M.2 2280 form factor. We document the design-space exploration that led to the 4 mm² target, the LPDDR5X-vs-LPDDR4X PHY tradeoff that gates the model-class commitment, and an explicit pre-RTL audit of the spec that corrected eight numerical inconsistencies before any high-level synthesis work begins. Lambda is, to our knowledge, the first open-source academic standalone transformer accelerator targeting the 3–5 B-parameter on-device deployment regime.

Index Terms—LLM inference accelerator, KV-cache compression, TurboQuant, FlashAttention, paged attention, adaptive precision, TSMC N16FFC, academic ASIC, IMEC mini@sic

I. Introduction

The 3–5-billion-parameter transformer has become the practical on-device deployment regime for large language models in 2025–2026. Llama-3.2-3B [1], Qwen2.5-3B [2], Phi-3.5-mini [3], Ministral-3-3B [4], and Gemma-2-2B all

target this band; Apple’s recently-published foundation-language-model family [5], Google Gemini Nano, and Microsoft Copilot+ all deploy at similar scales. The accelerators that serve these workloads in production—Apple’s Neural Engine, Qualcomm’s Hexagon NPU—are proprietary integrated SoCs whose internal architectures are not publicly documented. Modern GPU accelerators [6]–[8] serve the workload at the data-center scale but at die areas and power envelopes 10²–10³ times Lambda’s. No open-source academic standalone accelerator at this model class and process node currently exists.

Lambda fills that gap. We present a 4 mm² ASIC at TSMC N16FFC, designed for tape-out through the IMEC / Europractice mini@sic 2.0 academic shuttle program [9] at a published 2026 academic shuttle price of €103,632 (approximately \$112–120 K) for the 4 mm² block. The chip runs a 3-billion-parameter W4A8 transformer at 6.3–7.7 decode tokens per second in a 2.6 W typical envelope, holding 4–8 GB of weights in an off-chip LPDDR5X-8533 x16 package and exposing host I/O over a PCIe Gen3 x1 link on an M.2 2280 carrier card.

Three architectural commitments differentiate Lambda from prior open-source accelerator designs:

(1) Rotation-codebook KV compression in silicon. Lambda’s KV-compression engine (KCE-mini) implements a rotation-codebook quantizer inspired by TurboQuant [10]—a 2025 algorithm combining a random orthogonal rotation, per-coordinate optimal scalar (Lloyd–Max) quantization, and a quantized-JL residual corrector—in a 0.08 mm² block that operates without multipliers on either the encode or decode path. At our 16-element block size, the effective rate is 4.0 bits per element, a 4.0× reduction versus FP16. We state the epistemic status plainly: TurboQuant’s published quality-neutral results [10] use a QR-based random orthogonal rotation over the full head dimension with two-tier channel-wise bit allocation at 3.5 bits per channel, and no per-group scale factors; KCE-mini’s 16-element block-codebook structure is a different

quantizer, and quality parity at 4.0 bpe is a design target to be validated in the pre-HLS golden-model evaluation, not yet an empirical result. We believe Lambda is the first silicon implementation of a TurboQuant-style rotation-codebook KV compressor at a competitive process node.

(2) Compressed-domain attention scoring. Because the rotation stage (a Walsh–Hadamard butterfly in KCE-mini’s restructuring) spreads outlier coordinates across all dimensions, the attention dot product $Q \cdot K^\top$ can be computed directly on the 3-bit compressed codebook indices and rescaled once at the row sum. Lambda’s matrix engine (MatE) therefore needs no FP16 multiplier datapath; an $\text{INT8} \times \text{INT3}$ multiplier with an INT24 K-axis accumulator suffices for attention scoring against the full KV cache. This eliminates approximately 0.4 mm^2 of FP16 fabric that comparable mixed-precision designs (e.g., [11]) require.

(3) Token Importance Unit (TIU) for adaptive precision. Inspired by the entropy-driven KV quantization analysis of [11], we introduce a small (0.03 mm^2) on-die accumulator that tracks the cumulative softmax weight per 16-token KV block. The TIU output drives two consumer paths: a heavy-hitter eviction policy in the memory subsystem controller (H2O-style retention [12]) and a per-block precision selector that lets the KCE downgrade low-importance blocks to 3.0 or 2.0 bits per element. To our knowledge, this is the first silicon implementation of adaptive-precision KV cache management; published prior work is software-only.

The remainder of this paper develops these contributions in detail. Section II surveys the prior art the design is grounded in. Section III describes Lambda’s seven functional blocks and their top-level grouping. Section IV traces the decode token data flow and the quantization decisions that shape it. Section V reports the area and power budgets, including a pre-RTL audit that corrected eight numerical inconsistencies in the original spec. Section VI characterizes Lambda’s coverage across the candidate-model space and quantifies the load-bearing LPDDR5X-vs-LPDDR4X PHY tradeoff. Section VII discusses what we deliberately excluded and why.

II. Background and Related Work

Lambda integrates four lines of recent research into a single chip target.

Online attention. FlashAttention [13]–[15] reformulates softmax as an online tiled reduction with per-row running maxima m_i and running sums-of-exponentials ℓ_i . This eliminates materialization of the score matrix and reduces the activation-memory pressure that dominates long-context attention. Lambda’s vector unit (VecU) implements the FlashAttention-3 online softmax algorithm in microcode; the running state per row is two FP16 scalars plus the running output accumulator O_i . VecU also implements the rotary positional embedding scheme

(RoPE) [16], RMSNorm [17], and the SwiGLU gating used by the Llama-3 family [18] as microcoded loops.

Paged KV management. vLLM’s PagedAttention [19] treats the KV cache as a virtual address space with fixed-size physical blocks, indexed by a per-sequence block table. The pattern enables zero-fragmentation memory use and dynamic eviction. Lambda’s memory subsystem controller (MSC) implements this in silicon as a 128-entry content-addressable block table with 16-token blocks, the silicon analog of vLLM’s logical-to-physical mapping. FlashInfer [20] extends the paged abstraction to a composable taxonomy of attention patterns (dense paged, sparse structured, ragged, composite); Lambda’s MSC exposes a CSR-selectable mode covering this taxonomy.

Weight and activation quantization. Large-scale evaluation shows 4-bit weight quantization recovers 96–99% of unquantized accuracy across model scales [21] (that study’s headline configurations are W4A16, W8A8, and FP8). We pair W4 weights (via GPTQ [22] or AWQ [23]) with 8-bit per-token dynamic activations (via SmoothQuant [24]), a common deployment pairing for the 3–5 B class. Lambda’s MatE multiplies INT8 activations against INT4 weights as its primary path.

KV-cache compression. A growing body of work [10], [25]–[28] compresses KV caches via uniform quantization, rotation, codebook quantization, or per-token mixed precision. TurboQuant [10] applies a random orthogonal rotation (QR decomposition of a Gaussian matrix) over the full head dimension, concentrating coordinates toward a near-Gaussian distribution, followed by per-coordinate optimal scalar (Lloyd–Max) quantization with two-tier channel-wise bit allocation (in its 2.5-bit setting, 32 outlier channels at 3 bits and 96 at 2 bits for $d=128$) and a 1-bit quantized-JL residual corrector—reaching quality-neutral KV compression at 3.5 bits per channel ($4.57\times$ vs. FP16) with no per-group scale factors. The paper reports no experiments below $d=200$, and its concentration arguments rely on high dimension. Lambda’s KCE-mini adopts the rotation-plus-codebook recipe but restructures it as a 16-element block quantizer with a uniform 3-bit codebook and per-block scale, yielding 4.0 bits per element ($4.0\times$): a different quantizer whose quality at 4.0 bpe is unvalidated and is gated on the pre-HLS golden-model evaluation (Section VII). The rotation property is what motivates compressed-domain attention scoring at $\text{INT8} \times \text{INT3}$ without the FP16 fallback paths that adaptive-precision designs [11] otherwise carry; confirming that the property survives at our block size is part of the same pending evaluation.

Attention-weight-driven eviction and adaptive precision. H2O [12], Scissorhands [29], TOVA [30], and the StreamingLLM “attention-sink” phenomenon [31] all show that retaining tokens whose cumulative attention weight exceeds a threshold (and evicting the rest) preserves long-context quality at a fraction of the KV-cache footprint. The recent adaptive-precision-KV work [11] extends this

idea to bit-width allocation: high-attention tokens stay at higher precision, low-attention tokens demote. Lambda’s TIU is the silicon expression of this idea at per-block granularity (16 tokens per block, matching the PagedAttention block size).

III. Architecture

Figure 1 shows Lambda’s top-level decomposition. Seven functional blocks are grouped under five top-level units—ACU (Attention Compute Unit), MSC (Memory Subsystem Controller), LSU (Layer Sequencer), TIU (Token Importance Unit), and HIF (Host Interface)—plus 0.8MB of on-die SRAM in four banks and a vendor-IP LPDDR5X x16 PHY. The ACU is itself a composite of three blocks (MatE, VecU, KCE-mini) that share the compute fabric of the chip; the naming convention follows [32] while preserving the internal block structure for high-level synthesis (HLS) modularity. We discuss each block in turn.

A. Matrix Engine (MatE)

MatE is an 8×8 weight-stationary systolic array of 64 $\text{INT8} \times \text{INT4}$ processing elements (PEs). At 1 GHz, peak throughput is 128 GOPS; sustained throughput at 60% utilization is 76.8 GOPS—a figure that is, in the bandwidth-bound regime, $1.6 \times$ the throughput the chip needs at the largest model size. The factor is constant across model sizes in this regime because the GOPS demand scales with bandwidth (not with model size); we found and corrected an earlier draft claim of “3–5 $\times$ headroom” as a derivation error during the spec audit.

Two dataflow modes are CSR-selectable:

Weight-stationary for the GEMMs that dominate transformer compute: Q/K/V projections, attention output projection, FFN up- and down-projections, and the final logits projection. Weights pin once per tile; activations stream from the activation buffer; products accumulate down the columns.

Output-stationary for the $Q \cdot K^\top$ attention scoring step. Q vectors pin for the duration of the attention tile; compressed K values stream from the kv_scratchpad. Because K is stored in TurboQuant 3-bit codebook indices, MatE multiplies $\text{INT8} \times \text{INT3}$ directly—no dequantization to FP16 is required. The signed products (12 bits in the $(-128) \times (-8)$ corner) feed an INT24 reduction along the head-dimension axis; a single FP16 dequantization scale is applied once at the row sum. We selected INT24 (rather than the INT16 the original spec contained) after analysis showed that reducing $K=128$ $\text{INT8} \times \text{INT4}$ products can reach a magnitude of $1024 \times 128 = 2^{17}$, requiring at least 18 signed bits—outside INT16’s range. INT24 is itself not unconditionally safe: at the FFN down-projection’s $K=8192$, the adversarial all-corner sum $(-128 \times -8) \times 8192 = 2^{23}$ exceeds the INT24 maximum by exactly one. A planned numerical sweep over the demo models’ per-layer dimensions will either confirm this

corner is unreachable under W4A8 value distributions or trigger the named INT26 fallback width. The INT16 width survives only as the partial-product register inside each PE, following the per-PE/per-column accumulator split of Jouppi’s TPU [33], [34].

B. Vector Unit (VecU)

VecU is eight FP16/BF16 SIMD lanes sharing a 1K-instruction microcode RAM and three 64-entry transcendental LUTs (exp, rsqrt, sigmoid) with linear interpolation. The unit handles every non-GEMM operation in the transformer: FlashAttention-3 online softmax, rotary positional embedding (RoPE) pair-rotation, RMSNorm, SiLU/GELU, residual addition, and sampling. We chose programmable SIMD over fixed-function blocks for verification economy—one microcoded block has a smaller verification surface than four fixed-function blocks.

The online softmax microcode is the most consequential program. Per attention row, VecU maintains a triple (m_i, ℓ_i, O_i) : running max, running sum-of-exponentials, and running output accumulator. As each tile of scores arrives, VecU computes the new max, rescales the running accumulators by $e^{m_{\text{old}} - m_{\text{new}}}$, exponentiates the tile via the exp LUT, and accumulates into ℓ_i and into O_i weighted by the corresponding V tile. The 8 lanes parallelize across rows; tile size is 32–64 scores; approximately 32 microinstructions execute per tile.

C. KV Compression Engine (KCE-mini)

KCE-mini is the headline research IP. At 0.08 mm^2 , it implements TurboQuant compression and decompression on 16-element groups via three stages: (i) a 16-point Walsh–Hadamard butterfly (4 stages of 8 add/subtract operations each, 64 operations total) which rotates the input into a basis where outlier coordinates are spread roughly uniformly; (ii) an 8-centroid Lloyd–Max nearest-centroid classifier (7 comparators per lane $\times$ 16 lanes = 112 comparators) which assigns each rotated coordinate to a 3-bit codebook index; and (iii) bit-packing with a 16-bit FP16 per-group magnitude scale. The effective rate is $(16 \times 3 + 16)/16 = 4.0$ bits per element—a $4.0 \times$ compression versus FP16.

Critically, neither the encode nor the decode path contains a multiplier. The Hadamard butterfly is sign-pattern add/subtract; the codebook classifier is comparison-based; the decode path is a LUT read plus an inverse Hadamard. This is what allows KCE-mini to fit in 0.08 mm^2 at our node. Five CSR-selectable modes provide design flexibility: the primary TurboQuant 3-bit + 16-point mode, a linear Hadamard-INT4 fallback, an asymmetric K-3-bit/V-2-bit production mode that yields $4.57 \times$ compression on average, an FP4 (E2M1) codebook mode using NVFP4 levels [8], [35] or OCP Microscaling MXFP4 [36] via an alternative ROM, and an FP16 bypass for ablation.

We caught and corrected three inconsistent claims about KCE-mini’s compression rate during the pre-RTL audit.

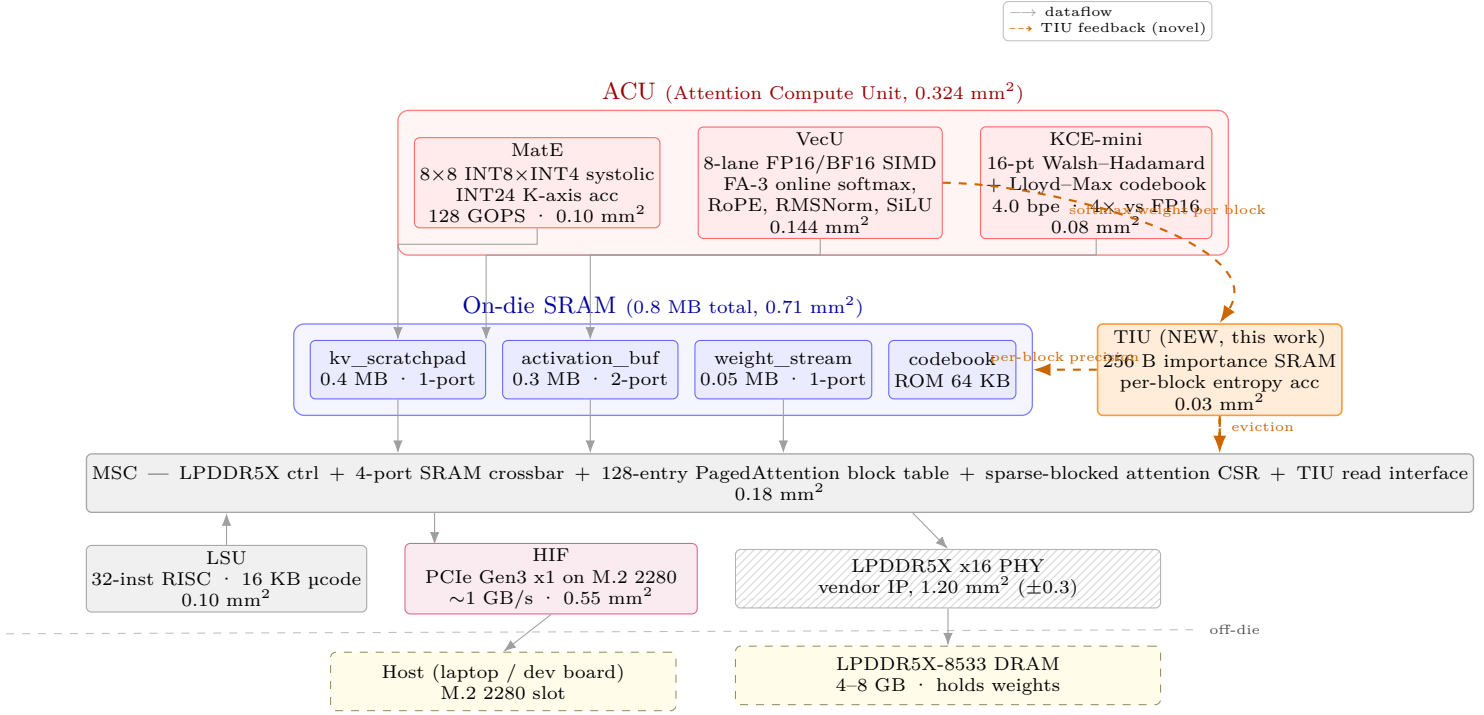


Fig. 1. Lambda’s top-level architecture, drawn full page width to keep the dataflow legible. The chip’s seven functional blocks are grouped under five top-level units. The ACU (Attention Compute Unit, red) contains the three compute blocks that share the GEMM and SIMD fabric (MatE, VecU, KCE-mini). The four on-die SRAM banks (blue, 0.8MB total) sit between the compute fabric and the memory controller. The MSC manages LPDDR access, the 128-entry PagedAttention block table, sparse-blocked attention CSR modes, and the TIU read interface. The LSU walks a pre-compiled 16 KB microcode schedule. The TIU (orange, novel to this work) accumulates per-block attention entropy from VecU’s softmax broadcasts and drives two feedback paths drawn in dashed orange: per-block adaptive precision to KCE-mini, and heavy-hitter-driven eviction to MSC. The HIF is a PCIe Gen3 x1 endpoint on an M.2 2280 carrier; the LPDDR5X x16 PHY drives the off-die DRAM package. Areas shown are at the PHY mid-case (1.20 mm^2).

Earlier spec drafts variously claimed 3.5 bits/element (the flagship 32-point Hadamard rate), 5.3 bits/element (apparently including unspecified extra overhead), and $3.0\times$ compression. The 4.0 bits/element figure derives directly from first principles for the 16-point variant and is the canonical value.

D. Memory Subsystem Controller (MSC)

MSC integrates four functions in 0.18 mm^2 : the LPDDR5X x16 protocol-side controller (with open-page policy and bank-conflict avoidance), a 4-port SRAM crossbar serving MatE, VecU, KCE, and the host, a 128-entry block table providing the silicon analog of vLLM’s PagedAttention [19] indexing (16 tokens per block, 2K addressable tokens per session), and a DMA descriptor FSM. Sparse-blocked attention is exposed as a CSR-selectable mode (`attention_pattern_modes`) following the FlashInfer taxonomy [20]: dense paged, sliding window, sparse blocked, local+global hybrid (the Gemma-2 layout).

MSC is single-session by design. We exclude continuous batching (multi-session state), tier-3 host-DRAM eviction (we have no PCIe-to-host swap path on a 4 mm^2 die), and cross-session prefix sharing. These omissions are deliberate:

the chip is sized for batch-1 on-device inference, not multi-tenant serving.

E. Layer Sequencer (LSU)

The LSU is a minimal three-stage in-order RISC: 32-bit fixed-width instruction format, 32 instructions, 16×32 -bit general-purpose registers, 16 KB of microcode RAM holding the entire pre-compiled per-layer schedule, single-issue across three dispatch lanes (scalar, vector to ACU, DMA to MSC). No branch predictor, no out-of-order machinery, no cache hierarchy: transformer decode is structurally identical layer to layer, and a static schedule walked deterministically suffices. The host loads up to 4 K 32-bit microcode instructions at boot; the chip executes the same schedule forever. The footprint is 0.10 mm^2 .

F. Token Importance Unit (TIU)

TIU is a 0.03 mm^2 block introduced in this work. It maintains a 256-byte importance SRAM organized as 128 entries of 16 bits each (one entry per PagedAttention block). Update path: VecU broadcasts the per-block cumulative softmax weight of the current attention pass; the TIU adds to the corresponding importance register. Consumer paths: (i) MSC queries the lowest-importance block when the `kv_scratchpad` approaches

capacity (heavy-hitter eviction, H2O-style [12]), and (ii) KCE-mini queries the per-block importance bucket when re-compressing an evicted-and-recalled block, dropping low-importance blocks to 3.0 or 2.0 bits per element while keeping high-importance blocks at the 4.0 bpe primary mode.

Four CSR-selectable policies cover the design space: `tiu_off` (pure FIFO eviction, uniform precision), `tiu_h2o` (heavy-hitter retention, uniform precision), `tiu_streaming_llm` (recent + attention-sink retention), and `tiu_adaptive_precision` (heavy-hitter retention plus per-block adaptive precision). The block is grounded in the adaptive-precision-KV analysis of [11], which we believe has not previously been implemented in silicon.

G. Host Interface (HIF)

HIF is a PCIe Gen3 x1 endpoint (1 GB/s sustained) on the M.2 2280 form factor at the PCB level. The M.2 connector wires four PCIe lanes by specification; Lambda drives only one on-die, and PCIe link training negotiates the link down to x1. Vendor IP (Synopsys DesignWare or Cadence’s PCIe Gen3 PHY at 16 nm) provides the analog and protocol stack; we implement the BAR0 CSR map, the 16-deep doorbell command queue, and the JTAG TAP. JTAG runs on dedicated pins, separate from the PCIe lanes, so post-silicon debug remains accessible before PCIe link is up. Total block area is 0.55 mm², of which the vendor PHY accounts for approximately 0.35 mm².

We initially specified USB-C 2.0 for this interface and revised to PCIe Gen3 x1 mid-design after concluding that the 25-second weight-load latency of USB-C 2.0 was a development-iteration cost we did not want to accept, and that the M.2 form factor better matched the modern inference-card target. The PCIe vendor IP also has public 16 nm datasheets, in contrast to the LPDDR5X x16 PHY which remains NDA-gated at 16 nm; the PCIe-side risk profile is therefore lower.

IV. Dataflow and Quantization

We walk one decode token of a 3-billion-parameter model (Llama-3.2-3B, 28 layers, 8 KV heads of 128 dimensions, 3072 hidden width, FFN width 8192) through the pipeline to make the quantization commitments concrete.

The LSU issues approximately 200 microinstructions per layer. For each layer, the schedule alternates four operand-stationary patterns. (1) Q/K/V projection: MatE consumes the activation vector and three weight matrices (streamed from LPDDR via the `weight_stream_buffer`), in weight-stationary mode, producing a 3072-dimensional Q vector (24 query heads $\times$ 128) and 1024-dimensional K and V vectors (8 KV heads $\times$ 128 each). (2) RoPE rotation: VecU rotates Q and K pair-wise using the rotary frequency table. (3) KV compression: KCE-mini compresses the new K and V into 16-element groups, writing 3-bit codebook indices plus per-group FP16 scales into the `kv_scratchpad`. (4) Attention scoring: MatE

switches to output-stationary mode and computes $Q \cdot K^T$ against the compressed K values directly. The multiplier path is INT8 (Q) $\times$ INT3 (codebook index), the partial product is 12-bit signed (the PE register is sized for the INT8 $\times$ INT4 (−128) $\times$ (−8) corner shared with weight-stationary mode), the K-axis reduction is INT24, and the per-group FP16 scale is applied once at the row sum. K is never expanded back to FP16.

The scores feed VecU, which executes the FlashAttention-3 online softmax microcode in tiles of 32. Simultaneously, the running output accumulator O_i is updated by multiplying scaled scores against the (compressed) V values through the same MatE path. When the last tile is consumed, O_i is the attention output. The TIU has been quietly accumulating the per-block cumulative softmax weight throughout this loop; the importance SRAM is now ready for the next eviction or precision-tuning decision.

The post-attention path runs through MatE again (attention output projection, FFN up-projection, FFN down-projection) and VecU (SiLU, residual add, RMSNorm). End-to-end, the schedule completes one layer in approximately 4.8 ms at Llama-3.2-3B’s 1.605 GB W4 footprint on LPDDR5X x16; 28 layers compose a 134.4 ms-per-token decode cadence, or 7.4 tokens per second. Compute is bandwidth-bound throughout: the limiting factor is the time required to stream the weights of each linear layer from LPDDR.

A. Why no FP16 multiplier in MatE?

A common alternative approach—adopted, for example, in adaptive-precision-attention designs [11], [32]—routes “outlier” attention tiles to an FP16 multiplier and “inlier” tiles to an INT8 multiplier, with a per-tile gating function deciding which path to take. The design rationale is that some attention tiles are dominated by a few high-magnitude scores that would lose precision under INT8 quantization.

The rotation stage makes this gating unnecessary. The rotation spreads outlier coordinates across all dimensions, producing a roughly uniform distribution. After rotation, every coordinate is amenable to the same 3-bit codebook quantization; no tile is an “outlier” tile in the relevant sense. The published empirical evidence [10] is, however, at a different operating point than ours: quality-neutral results at 3.5 bits per channel using a QR-based random orthogonal rotation over the full head dimension with two-tier channel-wise bit allocation and no per-group scales. KCE-mini’s 16-element block-codebook variant at 4.0 bpe inherits the mechanism but not the measurements; confirming the property at our structure is the pre-HLS golden-model evaluation that gates the demo-model decision. We rely on the rotation property to omit the FP16 multiplier from MatE entirely, saving approximately 0.4 mm² of fabric area at our node. An FP16 path remains

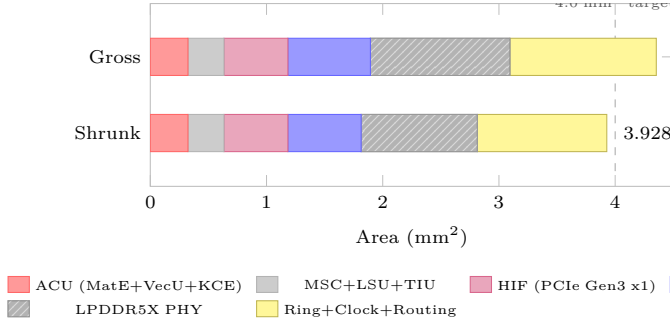


Fig. 2. Area accounting at PHY mid-case (Gross, 4.354 mm²) and after the recommended shrink path—vendor PHY at 1.0 mm² best case, I/O ring tightened from 100 to 80 μ m (−0.146 mm² by the spec’s ring-area formula; an earlier −0.06 figure was not reproducible and was corrected), activation buffer trimmed to 0.2 MB—which lands at 3.928 mm², 0.072 mm² under the 4.0 mm² target. Earlier spec drafts dropped the routing-overhead buffer (0.10 mm²) from the total and claimed 3.974 mm²; the pre-RTL audit corrected this.

in VecU, where it is needed for softmax, RMSNorm, RoPE, and other transcendental-heavy operations.

V. Implementation

A. Process, target, and shuttle

Lambda targets TSMC’s N16FFC FinFET process at a logic density of approximately 28.2 MTr/mm² and an HD SRAM density of approximately 1.25 MB/mm², with a core voltage of 0.8 V and a target clock of 1 GHz (800 MHz fallback specified). Tape-out is via the IMEC / Europractice mini@sic 2.0 academic shuttle program [9]; the Muse Semiconductor US-focused TSMC University FinFET path is the documented fallback. Shuttle cost at the published Europractice 2026 academic price (€26,592 for the first mm² plus €2,568 per additional 0.1 mm²) is €103,632 for the 4 mm² block, approximately \$112–120 K; total chip cost including PHY IP licensing (LPDDR5X x16 vendor PHY plus PCIe Gen3 x1 vendor PHY) is approximately \$225–310 K. The design flow is Cadence end-to-end: Stratus HLS for C++-to-RTL synthesis, Genus for logic synthesis, Innovus for place-and-route (with DREAMPlace [37] explored as an open-source comparison pre-tape-out), Pegasus for DRC/LVS, and Tempus (SSV) for static timing signoff. Compiler-side experiments target MLIR [38] as the lowering substrate for the LSU schedule generator. Tool access is bundled with mini@sic registration.

B. Area and power budgets

Table I reports per-block area, verification surface, and risk classification; Figure 2 visualizes the gross-versus-shrunk accounting. The gross total at the PHY mid-case (1.20 mm²) is 4.354 mm², exceeding the 4 mm² target by 0.354 mm². The recommended shrink path—PHY landing at 1.0 mm² vendor best case (−0.20 mm²), I/O ring tightened from 100 μ m to 80 μ m (−0.146 mm², recomputed from the spec’s ring-area formula; an earlier draft’s −0.06 figure could not be reproduced and was

TABLE I
Per-block area, verification surface, and risk class at PHY mid-case (1.20 mm²).

Block	Area (mm ²)	Tests	Risk
MatE (8×8 INT8×INT4)	0.10	50	medium
VecU (8-lane SIMD)	0.144	25	medium
KCE-mini (16-pt Hadamard)	0.08	60	low
MSC (LPDDR ctrl + xbar + BT)	0.18	50	medium
LSU (32-inst RISC)	0.10	30	low
TIU (256B + accumulator)	0.03	15	low
HIF (PCIe Gen3 x1)	0.55	30	medium
SRAM (0.8 MB, 4 banks)	0.71	—	low
LPDDR5X x16 PHY (vendor)	1.20	—	high
Clock + power grid	0.40	—	—
Routing overhead	0.10	—	—
I/O ring + pads + ESD	0.76	—	—
Gross total	4.354	260	
With recommended shrinks	3.928	—	

corrected), and the activation buffer shrunk from 0.3 MB to 0.2 MB (−0.08 mm²)—lands at 3.928 mm², 0.072 mm² (1.8%) under target. The shrink is contingent on the LPDDR5X x16 PHY vendor quote returning at or below 1.0 mm²; we discuss the LPDDR4X fallback in Section VI.

Power decomposes to 2.6 W typical decode and 3.3 W peak prefill at 1 GHz. The LPDDR5X PHY and DRAM together draw 0.72 W (at the 7.5 mW/Gbps planning figure for LPDDR5X-class memory and 96 Gbps sustained). MatE at 50% utilization is 0.32 W (at the 5 W/TOPS heuristic for INT8 systolic at 16 nm), VecU 0.16 W, SRAM 0.16 W, KCE 0.05 W, TIU 0.01 W, MSC+LSU+HIF 0.50 W. Whole-chip leakage at 16 nm at 0.8 V is approximately 0.70 W. The M.2 2280 slot delivers approximately 10 W per spec; we sit at 26% of that envelope at typical decode.

C. Pre-RTL audit

Before any HLS work begins we ran an explicit numerical audit of the spec and corrected eight inconsistencies. We document the audit here because each correction is a load-bearing decision that the rest of the chip integrates over.

KV bytes per token were 2.3× understated. The bytes-per-token-per-layer formula in the original spec used $k_h \times d_h \times \text{bpe}/8$, dropping the factor of 2 for K and V separately, and used the flagship 3.5 bpe figure rather than our 16-point variant’s 4.0 bpe. We re-derived every capacity number against the correct $k_h \times d_h \times 2 \times \text{bpe}/8$ formula at 4.0 bpe.

The “Qwen2.5-3B 32K on-die” claim was wrong. With corrected math, no 3-billion-parameter target model holds 32 K context fully on-die per layer; the 0.4 MB scratchpad holds 1638 tokens for the 2-KV-head Qwen layout and 409 tokens for the 8-KV-head Llama/Mistral layouts. Long contexts serve via LPDDR streaming, at an overhead of approximately 24 ms per token at 32 K for the Qwen layout

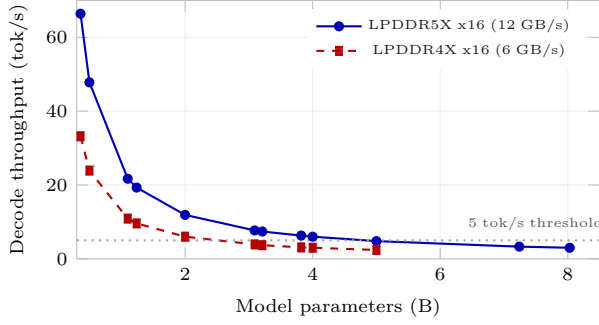


Fig. 3. Decode throughput versus model size under the two PHY options. Lambda’s baseline LPDDR5X x16 path serves the 3–5B-parameter class above the 5 tok/s threshold (Llama-3.2-3B at 7.4, Qwen2.5-3B at 7.7, Phi-3.5-mini at 6.3 tok/s). The LPDDR4X fallback path caps at approximately 2.4B parameters above threshold, recovering the 1–2B-parameter class (Llama-3.2-1B, Qwen2.5-1.5B, Gemma-2-2B).

(still above the 5 tok/s threshold) and approximately 75 ms per token for the 8-head layouts (below threshold).

MatE’s INT16 accumulator was a real bug. The original spec specified INT16 K-axis accumulators “because 3–5B models tolerate it.” $\text{INT8} \times \text{INT4}$ produces a signed product needing 12 bits in the $(-128) \times (-8)$ corner; reducing $K=128$ requires 18 signed bits, saturating INT16 after as few as 32 worst-case accumulations. Corrected to INT24 with an INT16 partial-product register inside each PE; INT24’s own residual $K=8192$ corner (exactly INT24 max + 1) is documented in Section III with the INT26 fallback. The model class does not bear on the accumulator width.

The “3–5× compute headroom” claim was a derivation error. In the bandwidth-bound regime, $\text{GOPS}_{\text{needed}} = 2 \cdot \text{BW}_{\text{GB/s}} / \text{bytes}_{\text{per param}}$, a constant across model sizes (because both numerator and denominator scale with bandwidth). The correct figure is a constant 1.6× headroom.

Additional corrections: the “comfortable headroom at PHY = 1.0 mm²” claim was misleading (the sensitivity sweep actually returns +0.002 mm² at 1.0 MB SRAM—exactly fits, no margin); the weight_stream_buffer was 0.05/0.15 MB inconsistent (canonical 0.05); the total SRAM was 0.8/0.85/1.0 MB inconsistent (canonical 0.8); the KCE-mini effective rate was 3.5/4.0/5.3 bpe inconsistent (canonical 4.0).

None of the corrections changed any architectural decision. Every fix was either a math error or a stale claim from an earlier iteration. We document them here because pre-RTL audits of this kind are, in our experience, frequently skipped in academic chip projects and frequently catch real bugs.

VI. Performance and the LPDDR PHY Tradeoff

Decode is bandwidth-bound at every reasonable model size. The largest model the chip can serve at a given throughput threshold derives directly from the bandwidth: $\text{params}_{\text{max}} = \text{BW}_{\text{GB/s}} \cdot t_{\text{per tok}} / \text{bytes}_{\text{per param}}$. At

LPDDR5X x16 (12 GB/s sustained, $t = 200$ ms for 5 tok/s, 0.5 B/param for W4), the ceiling is 4.8 B parameters. Figure 3 shows the decode throughput curve across the candidate model set; the comfortable interactive zone (6.3–7.7 tok/s) covers Llama-3.2-3B (7.4 tok/s at its 1.605 GB W4 footprint and 134.4 ms/token—the primary demo target), Qwen2.5-3B (7.7), and Phi-3.5-mini (6.3). The 5-billion-parameter point sits exactly at threshold; 7–8-billion-parameter models drop below. Ministral-3-3B [4] (Apache-2.0, December 2025) is a candidate addition to the demo set whose architecture configuration we have not yet verified against Lambda’s memory budget; we attach no performance numbers to it. Model licensing is compatible with an academic demo across the set: Llama-3.2 ships under the Llama 3.2 Community License (attribution and derivative-naming passthrough), Qwen2.5-3B under the Qwen Research License (non-commercial), Phi-3.5-mini under MIT, and Ministral-3-3B under Apache-2.0.

A. The LPDDR5X-vs-LPDDR4X PHY tradeoff

The dominant uncertainty in Lambda’s area budget is the LPDDR5X x16 PHY footprint at 16 nm, which is NDA-gated and which we estimate at 1.20 mm² with ± 0.3 mm² real swing. The estimate is extrapolated from vendor x64 PHY data points; an authoritative number requires a written Synopsys or Cadence quote in Q2 2026. If the quote returns above 1.35 mm², Lambda’s recommended shrink path no longer closes at 4 mm², and a fallback is required.

The natural fallback is LPDDR4X x16 with a Cadence PHY. LPDDR4X has been shipping at 16 nm in multiple commercial parts since approximately 2018 and has public datasheets, in contrast to LPDDR5X’s NDA-only 16 nm references. The Cadence PHY for LPDDR4X at 16 nm is approximately 1.0 mm² with substantially less area variance (± 0.15 mm²). Adopting LPDDR4X frees approximately 0.2 mm² of breathing room and materially de-risks the first-FinFET-PHY integration that is the chip’s largest non-architectural risk.

The cost is bandwidth. LPDDR4X x16 sustains approximately 6 GB/s, halving the largest serviceable model class from approximately 4.8 B parameters to approximately 2.4 B at the 5 tok/s threshold. Llama-3.2-3B and Qwen2.5-3B drop below threshold; Llama-3.2-1B (9.6 tok/s), Qwen2.5-1.5B (8.0 tok/s), and SmolLM2-1.7B (7.1 tok/s) remain above it, with Gemma-2-2B (6.0 tok/s) borderline. The chip’s research thesis shifts from “first open-source academic standalone 3–5 B accelerator at 16 nm” to “first open-source academic standalone 1–2 B accelerator at 16 nm, with fully public-datasheet PHY IP”—a defensible alternative claim at materially lower tape-out risk.

The recommendation in our current planning is to commission both PHY quotes in parallel and gate the decision on the data. If the LPDDR5X x16 quote returns

at or below 1.25 mm^2 and a senior FinFET PD engineer is in place, the LPDDR5X path is correct; if the quote exceeds 1.35 mm^2 or the PD-engineer prerequisite is not met, the LPDDR4X path is correct. Neither outcome kills the project.

B. Comparison with prior art

Table II positions Lambda against the closest reference points we are aware of. The mobile-NPU comparison is necessarily qualitative because the Apple Neural Engine [5] and Qualcomm Hexagon NPU architectures are proprietary. Etched.ai’s Sohu transformer ASIC [39] sits at data-center scale on a substantially larger die. The academic-FPGA comparison reflects projects that target similar workloads at substantially larger die areas or older process nodes. Lambda’s distinguishing positions are (i) standalone operation at 3–5B model class without a tightly-coupled SoC, (ii) silicon-level TurboQuant, (iii) adaptive-precision KV via the TIU, and (iv) open-source academic distribution.

VII. Discussion

A. What we deliberately excluded

Lambda omits several mechanisms that are present in software-side frontier work. We document the exclusions because the omissions are deliberate and reversible in a future revision.

Multi-head Latent Attention (MLA). DeepSeek-V2’s MLA [40] compresses Q/K/V into a low-rank latent space before the head split, reducing per-layer KV by approximately an order of magnitude versus standard MHA. Lambda’s MSC and KCE-mini both assume the conventional $k_h \times d_h$ KV layout. MLA would require a redesign of both blocks; we defer it for a future revision and document the gap explicitly.

Continuous batching and prefix sharing. The single-session chip excludes multi-tenant serving primitives. We exclude them deliberately: a 4 mm^2 die does not have area for the multi-session state, and the on-device deployment target does not benefit from them.

Mixture-of-Experts. MoE routing requires a sparse-dispatch fabric that does not fit our area budget and that is not yet a dominant pattern in the 3–5B class.

Mamba and other SSM blocks. Selective state-space models [41] require different primitive operations from attention; the chip is not architected for them.

Multimodal vision/audio encoders. Out of scope for the chip’s text-only inference target.

Hardware speculative-decoding fabric. Speculative decoding (Medusa [42], EAGLE [43]) and multi-token prediction (DeepSeek-V3 [44]) are architecturally compatible: the host coordinates draft tokens, and the chip runs the verifier pass through a standard batched- N decode schedule. We do not include a dedicated speculative-decoding accelerator on-die; the LSU schedule is general enough to accommodate the pattern.

B. IP clearance: the Etched patent

Etched.ai’s US patent 2024/0419516 A1 [45] describes an AI hardware platform comprising two structurally separated circuits on a single IC: a systolic array for operations that do not consume past-token data and a self-attention circuit for operations that do. Lambda’s MatE is a single 8×8 systolic that mode-multiplexes between weight-stationary (for the no-past-token-data operations the patent describes as belonging to its systolic array) and output-stationary (for the past-token-data operations the patent describes as belonging to its self-attention circuit). The structural separation that grounds the patent’s claim is absent in Lambda’s design; the mode-switch is a CSR bit, not two circuits.

Our reading is that Lambda is outside the claim scope, but a defensive review with our institutional technology transfer office is the appropriate next step before silicon commit. The patent does highlight a real scheduling distinction we adopt explicitly: the LSU schedule treats weight-matmul tiles and attention-matmul tiles as separate dispatch primitives with separate optimization targets (K-axis size, operand stationarity).

C. Risks and open questions

The largest non-architectural risk is the LPDDR5X $\times 16$ PHY integration: this is the team’s first FinFET-era PHY tape-out, and the 1.20 mm^2 estimate has $\pm 0.3\text{ mm}^2$ real swing. Mitigation is the vendor quote (Q2 2026) and a senior FinFET physical-design engineer in place by Q3 2026; the LPDDR4X fallback (Section VI) is the documented release valve.

The IMEC mini@sic 2.0 published 2026 price for 4 mm^2 N16FFC is €103,632 (€26,592 for the first mm^2 plus €2,568 per additional 0.1 mm^2), approximately \$112–120K at 2026 exchange rates. A confirming quote via eptsmc@imec.be remains the first concrete action item on the project’s gantt.

VIII. Conclusion

We have presented Lambda, a 4 mm^2 open-source academic transformer-decode ASIC at TSMC N16FFC that serves the 3–5B-parameter on-device deployment regime at 6.3–7.7tok/s in 2.6W. The chip is, to our knowledge, the first silicon implementation of TurboQuant-style rotation-codebook KV compression at a competitive process node, the first silicon implementation of adaptive-precision KV cache management via on-die attention-entropy accumulation (the TIU), and the first open-source academic standalone accelerator targeting this model class. We have documented the design-space exploration, the LPDDR5X-versus-LPDDR4X PHY tradeoff that gates the model-class commitment, and an explicit pre-RTL audit that corrected eight numerical inconsistencies before any HLS work begins. The next phase of the project is high-level synthesis of the MatE processing element and the KCE-mini Hadamard butterfly—the two long

TABLE II
Lambda versus the closest accelerator reference points.

	Lambda (this work)	Apple ANE*	Qualcomm Hexagon NPU*	Academic FPGA accelerators
Process	TSMC N16FFC	≤N3	≤N4	16/28 nm
Die area	4mm ² (standalone)	integrated in SoC	integrated in SoC	100+ mm ² (FPGA fabric)
Model class @ 5 tok/s	3–5 B parameters	≥3 B	≥3 B	≤1 B
KV compression	TurboQuant 4.0 bpe	proprietary	proprietary	FP16 / INT8 (uniform)
Adaptive-precision KV	TIU (per-block)	unknown	unknown	none
Compressed-domain attention	yes (INT8×INT3)	unknown	unknown	no
Open source	yes	no	no	varies
Form factor	M.2 2280 carrier	integrated SoC	integrated SoC	dev board
Tape-out path	IMEC mini@sic 2.0	internal	internal	FPGA bitstream (no tape-out)

*Apple ANE and Qualcomm Hexagon NPU architectural details are proprietary; entries reflect publicly disclosed capabilities. Comparison is qualitative.

poles of the implementation effort. The full architecture specification, the audit log, the literature review, and the per-block HLS scaffolding are publicly available at github.com/LonghornSilicon/architecture.

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